

HYDROGENATED NITRILE RUBBER (HNBR) SYNTHESIS BY SIMULTANEOUS HYDROGENATION AND METATHESIS

1. ABSTRACT

This report synthesizes patent evidence on the preparation of hydrogenated nitrile rubber (HNBR), with emphasis on routes that reduce residual unsaturation while also lowering molecular weight and narrowing molecular weight distribution. The target material is hydrogenated nitrile rubber derived from nitrile rubber (NBR), including partially and highly hydrogenated variants, with improved heat, ozone, and oil resistance relative to the precursor. The two primary patents both focus on one-step or simultaneous hydrogenation/metathesis strategies rather than conventional two-step hydrogenation after separate chain scission. EP2473281B1 discloses a process for producing low molecular weight HNBR by reacting nitrile rubber in hydrogen in the presence of a ruthenium- or osmium-based catalyst capable of both metathesis and hydrogenation, optionally with a co-olefin. Its examples compare catalyst systems and show that the inventive catalyst can achieve faster reduction of residual double bond content while retaining manageable molecular weight and Mooney viscosity. US9079979B2 is closely related and similarly claims simultaneous hydrogenation and metathesis of nitrile rubber in hydrogen using catalysts of general formulae (I)–(III), again optionally with a co-olefin. Its examples compare Grubbs III, Grubbs II, and Wilkinson's catalyst, showing that Grubbs III provides faster hydrogenation than the other systems at comparable metal contents. Across the patent set, the dominant process trend is the use of ruthenium-based metathesis/hydrogenation catalysts to combine backbone scission and saturation in a single step, enabling low residual double bond content, reduced molecular weight, and narrower dispersity. The evidence also highlights the importance of catalyst identity, catalyst loading, and reaction time in controlling hydrogenation rate and final HNBR properties.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: EP2473281B1
- Patent Title: Process for the preparation of low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: EP

- Publication Year: 2014
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses a one-step process for producing hydrogenated nitrile rubber by simultaneously hydrogenating and metathesizing nitrile rubber in the presence of hydrogen. The process uses at least one ruthenium- or osmium-containing catalyst of the disclosed general formula and may optionally include a co-olefin, enabling concurrent chain scission and saturation to form low molecular weight HNBR. - Catalyst amount: 0.041 phr — Example 1, HNBR 1 (Hoveyda) - Catalyst amount: 0.045 phr — Example 1, HNBR 2 (Arlt) - Catalyst amount: 0.060 phr — Example 1, HNBR 3 (Wilkinson's) - Metal content: 65.6 ppm — Example 1, HNBR 1 - Metal content: 65.6 ppm — Example 1, HNBR 2 - Metal content: 65.6 ppm — Example 1, HNBR 3 - Co-catalyst: NIL — Example 1, HNBR 1 - Co-catalyst: NIL — Example 1, HNBR 2 - Co-catalyst: triphenylphosphine — Example 1, HNBR 3 - Co-catalyst amount: 1.0 phr — Example 1, HNBR 3 - Reaction time: 0 min — Example 1, Table 2 - Reaction time: 60 min — Example 1, Table 2 - Reaction time: 120 min — Example 1, Table 2 - Reaction time: 180 min — Example 1, Table 2 - Reaction time: 240 min — Example 1, Table 2 - Residual double bond content (RDB): 100% at 0 min — Example 1, Table 2 - Residual double bond content (RDB): 2.5% at 60 min for HNBR 1 — Example 1, Table 2 - Residual double bond content (RDB): 7.0% at 60 min for HNBR 2 — Example 1, Table 2 - Residual double bond content (RDB): 15.4% at 60 min for HNBR 3 — Example 1, Table 2 - Residual double bond content (RDB): 0.6% at 120 min for HNBR 1 — Example 1, Table 2 - Residual double bond content (RDB): 4.5% at 120 min for HNBR 2 — Example 1, Table 2 - Residual double bond content (RDB): 3.1% at 120 min for HNBR 3 — Example 1, Table 2 - Residual double bond content (RDB): 0.5% at 180 min for HNBR 1 — Example 1, Table 2 - Residual double bond content (RDB): 3.9% at 180 min for HNBR 2 — Example 1, Table 2 - Residual double bond content (RDB): 1.3% at 180 min for HNBR 3 — Example 1, Table 2 - Residual double bond content (RDB): 0.4% at 240 min for HNBR 1 — Example 1, Table 2 - Residual double bond content (RDB): 3.5% at 240 min for HNBR 2 — Example 1, Table 2 - Residual double bond content (RDB): 0.7% at 240 min for HNBR 3 — Example 1, Table 2 - Mw: 113371 g/mol — Example 1, Table 3, HNBR 1 - Mn: 48352 g/mol — Example 1, Table 3, HNBR 1 - PDI: 2.34 — Example 1, Table 3, HNBR 1 - Mooney viscosity: 33.3 — Example 1, Table 3, HNBR 1 - Mw: 133758 g/mol — Example 1, Table 3, HNBR 2 - Mn: 54595 g/mol — Example 1, Table 3, HNBR 2 - PDI: 2.45 — Example 1, Table 3, HNBR 2 - Mooney viscosity: 41.3 — Example 1, Table 3, HNBR 2 - Mw: 195304 g/mol — Example 1, Table 3, HNBR 3 - Mn: 64130 g/mol — Example 1, Table 3, HNBR 3 - PDI: 3.05 — Example 1, Table 3, HNBR 3 - Mooney viscosity: 69.3 — Example 1, Table 3, HNBR 3 - Mw range: 20,000 to 250,000 — Synthesis passage - Mooney viscosity range: 1 to 50 — Synthesis passage - MWD/PDI: less than 2.8 — Synthesis passage

Relevant Experimental Evidence - Example 1: Compared Hoveyda, Arlt, and Wilkinson's catalyst systems for simultaneous hydrogenation/metathesis and showed the inventive catalyst gave the fastest reduction in residual double bond content. - Example 1: Demonstrated that the inventive catalyst system produced low molecular weight HNBR with Mw 113371 g/mol and PDI

2.34. - Example 1: Showed that catalyst choice strongly affected final Mooney viscosity, with HNBR 1 at 33.3 versus HNBR 3 at 69.3. - Example 1: Confirmed that residual double bond content could be reduced to 0.4% after 240 min under the inventive conditions.

Technical Relevance This patent is directly relevant because it teaches a simultaneous hydrogenation/metathesis route to HNBR, which is a nontraditional but efficient way to reduce unsaturation and molecular weight in one step. It provides comparative catalyst data showing how catalyst selection controls hydrogenation rate, residual unsaturation, and final polymer properties.

Patent 2

- Patent Number: US9079979B2
- Patent Title: Process for the preparation of hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2015
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses simultaneous hydrogenation and metathesis of nitrile rubber in the presence of hydrogen, using catalysts selected from ruthenium- or osmium-based general formulae that can catalyze both transformations. The process optionally uses a co-olefin and is designed to produce hydrogenated nitrile rubber with lower molecular weight and narrower molecular weight distribution in a single step. - Catalyst amount: 0.057 phr — Example 1, HNBR 1 (Grubbs III) - Catalyst amount: 0.080 phr — Example 1, HNBR 2 (Grubbs III) - Catalyst amount: 0.06 phr — Example 1, HNBR 3 (Wilkinson's) - Catalyst amount: 0.055 phr — Example 1, HNBR 4 (Grubbs II) - Metal content: 65.6 ppm — Example 1, HNBR 1 - Metal content: 91.3 ppm — Example 1, HNBR 2 - Metal content: 65.6 ppm — Example 1, HNBR 3 - Metal content: 65.6 ppm — Example 1, HNBR 4 - Reaction time: 0 min — Example 1, Table 2 - Reaction time: 60 min — Example 1, Table 2 - Reaction time: 120 min — Example 1, Table 2 - Reaction time: 180 min — Example 1, Table 2 - Reaction time: 240 min — Example 1, Table 2 - Residual double bond content (RDB): 100% at 0 min for all HNBR samples — Example 1, Table 2 - Residual double bond content (RDB): 2.0% at 60 min for HNBR 1 — Example 1, Table 2 - Residual double bond content (RDB): 0.7% at 60 min for HNBR 2 — Example 1, Table 2 - Residual double bond content (RDB): 15.4% at 60 min for HNBR 3 — Example 1, Table 2 - Residual double bond content (RDB): 2.9% at 60 min for HNBR 4 — Example 1, Table 2 - Residual double bond content (RDB): 0.4% at 120 min for HNBR 1 — Example 1, Table 2 - Residual double bond content (RDB): 0.5% at 120 min for HNBR 2 — Example 1, Table 2 - Residual double bond content (RDB): 3.1% at 120 min for HNBR 3 — Example 1, Table 2 - Residual double bond content (RDB): 0.6% at 120 min for HNBR 4 — Example 1, Table 2 - Residual double bond content (RDB): 0.3% at 180 min for HNBR 1 — Example 1, Table 2 - Residual double bond content (RDB): 0.3% at 180 min for HNBR 2 — Example 1, Table 2 - Residual double bond content (RDB): 1.3% at 180 min for HNBR 3 —

Example 1, Table 2 - Residual double bond content (RDB): 0.3% at 180 min for HNBR 4 —
 Example 1, Table 2 - Residual double bond content (RDB): 0.6% at 240 min for HNBR 3 —
 Example 1, Table 2 - Mw: 191541 g/mol — Example 1, Table 3, HNBR 1 - Mn: 68643 g/mol —
 Example 1, Table 3, HNBR 1 - PDI: 2.79 — Example 1, Table 3, HNBR 1 - Mooney viscosity: 38.2
 — Example 1, Table 3, HNBR 1 - Mw: 135772 g/mol — Example 1, Table 3, HNBR 2 - Mn: 58259
 g/mol — Example 1, Table 3, HNBR 2 - PDI: 2.33 — Example 1, Table 3, HNBR 2 - Mooney
 viscosity: 20.2 — Example 1, Table 3, HNBR 2 - Mw: 195304 g/mol — Example 1, Table 3, HNBR
 3 - Mn: 64130 g/mol — Example 1, Table 3, HNBR 3 - PDI: 3.05 — Example 1, Table 3, HNBR 3 -
 Mooney viscosity: 69.3 — Example 1, Table 3, HNBR 3 - Mw: 144212 g/mol — Example 1, Table
 3, HNBR 4 - Mn: 59851 g/mol — Example 1, Table 3, HNBR 4 - PDI: 2.45 — Example 1, Table 3,
 HNBR 4 - Mooney viscosity: 32.6 — Example 1, Table 3, HNBR 4

Relevant Experimental Evidence - Example 1: Compared Grubbs III, Wilkinson's, and Grubbs II catalyst systems for simultaneous hydrogenation/metathesis of nitrile rubber. - Example 1: Showed Grubbs III catalysts gave rapid hydrogenation, with RDB dropping to 2.0% and 0.7% at 60 min for HNBR 1 and HNBR 2. - Example 1: Demonstrated that HNBR 2 achieved a low Mooney viscosity of 20.2 and PDI of 2.33. - Example 1: Confirmed that Wilkinson's catalyst was slower, retaining 15.4% RDB at 60 min and 0.6% at 240 min. - Example 1: Showed that catalyst identity and loading influence both hydrogenation rate and final molecular weight/viscosity.

Technical Relevance This patent is highly relevant because it discloses the same core HNBR manufacturing concept: simultaneous hydrogenation and metathesis of nitrile rubber in hydrogen. Its comparative example data help identify catalyst systems that improve hydrogenation efficiency while controlling molecular weight and dispersity.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- Both patents use a one-step simultaneous hydrogenation/metathesis strategy for nitrile rubber rather than a separate hydrogenation after chain scission.
- Both disclose ruthenium/osmium catalyst families capable of catalyzing both metathesis and hydrogenation, with optional co-olefin use.
- EP2473281B1 compares Hoveyda, Arlt, and Wilkinson's catalyst systems; US9079979B2 compares Grubbs III, Grubbs II, and Wilkinson's catalyst.
- In both patents, catalyst identity and loading are key variables controlling residual double bond content, molecular weight, and Mooney viscosity.
- EP2473281B1 reports catalyst amounts of 0.041, 0.045, and 0.060 phr at 65.6 ppm metal for the compared systems; US9079979B2 reports 0.057, 0.080, 0.06, and 0.055 phr with metal contents of 65.6, 91.3, 65.6, and 65.6 ppm.
- EP2473281B1 shows RDB falling to 0.4% by 240 min for HNBR 1, while US9079979B2 shows RDB reaching 0.3% by 180 min for HNBR 1 and HNBR 2.

- EP2473281B1 reports Mw values from 113371 to 195304 g/mol and Mooney viscosity from 33.3 to 69.3; US9079979B2 reports Mw values from 135772 to 195304 g/mol and Mooney viscosity from 20.2 to 69.3.
- US9079979B2 explicitly states that Grubbs III hydrogenates faster than Grubbs II or Wilkinson's catalyst at equal metal contents, whereas EP2473281B1 emphasizes the inventive catalyst's superior hydrogenation efficiency versus the industrial norm.

4. CONCLUSION

The patent evidence supports a clear synthesis direction for HNBR: simultaneous hydrogenation and metathesis of nitrile rubber in hydrogen using ruthenium-based catalysts that can perform both functions. The most effective conditions in the examples are those that combine appropriate catalyst identity with moderate catalyst loading, enabling rapid reduction of residual double bond content while producing low-to-moderate molecular weight polymers with controlled Mooney viscosity. Based on the evidence, catalyst systems in the Grubbs/Hoveyda family and related Ru complexes are preferred over Wilkinson's catalyst for integrated hydrogenation/metathesis performance. The reported data suggest that practical HNBR targets can be achieved in the range of Mw roughly 20,000 to 250,000, Mooney viscosity 1 to 50, and low residual unsaturation, with reaction time and catalyst selection being the dominant control variables.

5. REFERENCES

- EP2473281B1 | Process for the preparation of low molecular weight hydrogenated nitrile rubber | Not disclosed | EP | 2014 | <https://patents.google.com/patent/EP2473281B1/en>
- US9079979B2 | Process for the preparation of hydrogenated nitrile rubber | Not disclosed | US | 2015 | <https://patents.google.com/patent/US9079979B2/en>